

DSA4262 Group 2 (Group Project Report)

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1. Problem Canvas

1.1 - Problem Statement

NUS students experiencing depression, whether in their first episode or managing a relapse, struggle to receive timely and proportionate mental health support within the current UHC counselling framework. This is due to the absence of any between-session monitoring mechanism, meaning that fluctuations in mental state go undetected and unaddressed between appointments. This results in unmonitored symptom deterioration, elevated risk of relapse and an increased likelihood of reaching a crisis point before the students' next scheduled session.

1.2 - User Persona

To ground the problem in lived experience, a representative student persona was constructed based on the profile of NUS students currently enrolled in UHC counselling for depression, reflecting the structural gaps identified in the problem statement above. Mei Ling is a 21-year-old Year 3 NUS undergraduate who has been diagnosed with moderate depression. She is currently enrolled in UHC's counselling programme, with sessions scheduled at bi-weekly intervals of between 7 to 14 days. While these sessions provide a degree of structured therapeutic support, the intervals between appointments represent extended unmonitored periods during which her symptoms can fluctuate significantly, without any formal point of contact with her counsellor or the broader UHC support system.

Mei Ling's primary goal is to manage her depressive symptoms sufficiently to sustain her academic performance and maintain basic daily functioning. However, she faces a persistent and distressing challenge: when her condition deteriorates between sessions, there is no structured mechanism through which she can communicate this to her care team, and no clear clinical threshold to guide her in determining whether her experience warrants urgent outreach. The sessions, constrained by fixed scheduling, are often insufficient in frequency and depth to adequately address the evolving severity of her presentation, leaving critical gaps in her care. As she describes it: *"Some days, I spiral really fast, like within hours everything just crashes and I don't know what to do. I sit there thinking, do I call someone? I don't want to overreact but I feel like I'm completely falling apart and no one knows. The next session feels so far away. I once spent four days barely getting out of bed, crying for reasons I couldn't even explain. By the time I saw my counselor, I didn't even know how to describe it anymore. It feels like the system only sees me when I'm functional enough to show up, not when I actually need help the most."*

This example illustrates the central risk that the proposed system seeks to address. The absence of between-session monitoring, combined with the insufficient frequency of scheduled appointments, creates conditions in which symptomatic deterioration can escalate into a mental health crisis before clinical intervention is possible. Left undetected, such deterioration is associated with more severe depressive episodes, greater functional impairment and poorer long-term treatment outcomes, underscoring the need for a more continuous and responsive model of student mental health support.

1.3 - Jobs to Be Done (JTBD)

JTBD 1 (Early Warning Detection):

When my depressive symptoms worsen between counselling sessions, I want to detect early warning signs of deterioration through a structured check-in, so I can receive timely intervention before my condition escalates to a crisis point, without having to wait until my next scheduled appointment or bear the burden of self-identifying as being in crisis.

Design Rationale for JTBD 1:

Between-session gaps of 7 to 14 days create a significant monitoring blind spot in which students may experience a worsening depressive episode without any structured mechanism to communicate this to their care team. Without a reliable touchpoint during this interval, early warning signs of deterioration go undetected and students are left to navigate distress alone, often without knowing whether what they are experiencing warrants urgent contact. A PHQ-2 daily check-in directly addresses this by providing a low-friction, clinically standardised signal that captures real-time changes in symptom severity. By surfacing deterioration as it occurs rather than retrospectively at the next scheduled session, the check-in enables timely escalation of support, ensuring that students receive intervention when they actually need it, without requiring them to self-identify as being in crisis.

JTBD 2 (Receiving Adequate Care And Support At The Right Time):

When I am in a particularly difficult period between sessions, I want to receive care that truly meets what I am dealing with, so I can get through this period without having to wait for a scheduled appointment to access the support I need, thereby making it less likely for my condition to deteriorate further.

Design Rationale for JTBD 2:

The current fixed-interval scheduling model treats all students uniformly regardless of their actual clinical state, meaning that a student who is struggling significantly between sessions receives no more support than one who is coping well. This one-size-fits-all approach fails to reflect the dynamic and often unpredictable nature of depressive episodes, leaving students with elevated symptom severity without access to care proportionate to their level of need. The triage system addresses this directly by using PHQ-2 check-in data to identify high-severity cases in real time and escalate them for prioritised counsellor attention. This enables UHC counselors to allocate their limited availability to students who need it most urgently, ensuring that when a student is genuinely struggling, they are met with targeted interventive support rather than generic resources, delivering the right level of care at the right moment.

1.4 - Data-Driven Motivation

Three datasets were used to contextualise the problem and validate the intervention design.

1. IHME Global Burden of Disease: Depressive Disorders Dataset

This dataset provides nationally representative estimates of depression prevalence in Singapore, disaggregated by age group and sex, including both percentage rates and absolute case counts. Extraction of the 15 to 24 age band (corresponding to the university-aged population) reveals that this cohort is among the highest-prevalence demographic segments in Singapore, establishing that depressive disorder at this life stage is not incidental but a structural feature compounded by the academic and social pressures of university life. This directly motivates the selection of NUS students as the priority intervention population and grounds the project in the broader epidemiological reality of youth depression in Singapore.

2. Reddit & Twitter Depression Severity Levels Dataset

This dataset comprises Reddit and Twitter posts labelled by depression severity (mild, moderate and severe), capturing the linguistic and emotional patterns through which individuals express depressive symptomatology, including word choice, sentiment and emotional tone. It directly underpins the NLP model powering the chatbot triage layer, enabling the system to distinguish stable from escalating presentations by detecting varying degrees of depression in natural language without requiring direct counsellor involvement in every check-in interaction.

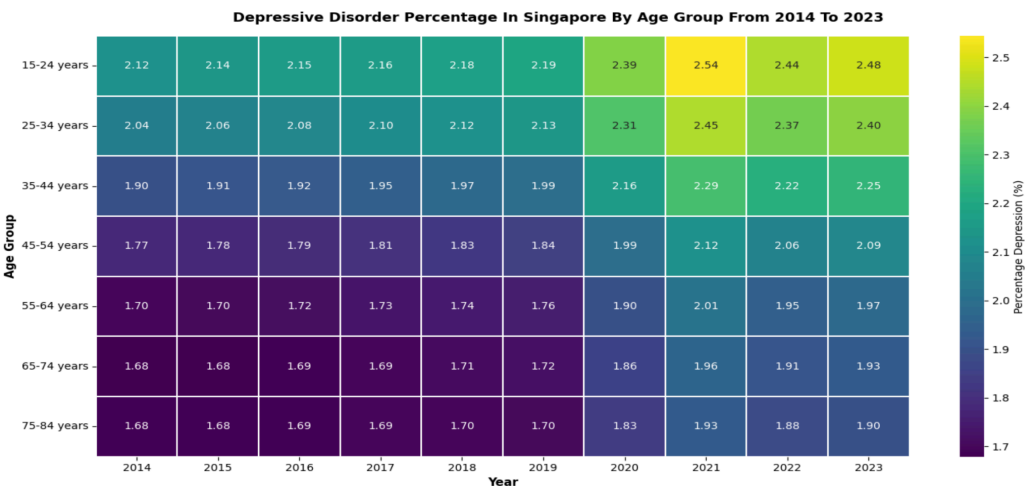
3. CBT Dataset (PHQ Scores + Session Data)

This clinical dataset captures patient experiences in CBT for anxiety and depression, encompassing PHQ scores recorded across multiple counselling touchpoints, consultation waiting times, session counts and session frequency. Analysis of PHQ score trajectories against session intervals provides the empirical basis for the project's central hypothesis that reducing the gap between symptom deterioration and professional response leads to measurably better clinical outcomes. Baseline patients drawn from this dataset will also serve as the reference cohort against which post-rollout PHQ delta comparisons are conducted, enabling rigorous evaluation of the system's clinical impact.

1.5 - Data Visualizations & Exploratory Data Analysis

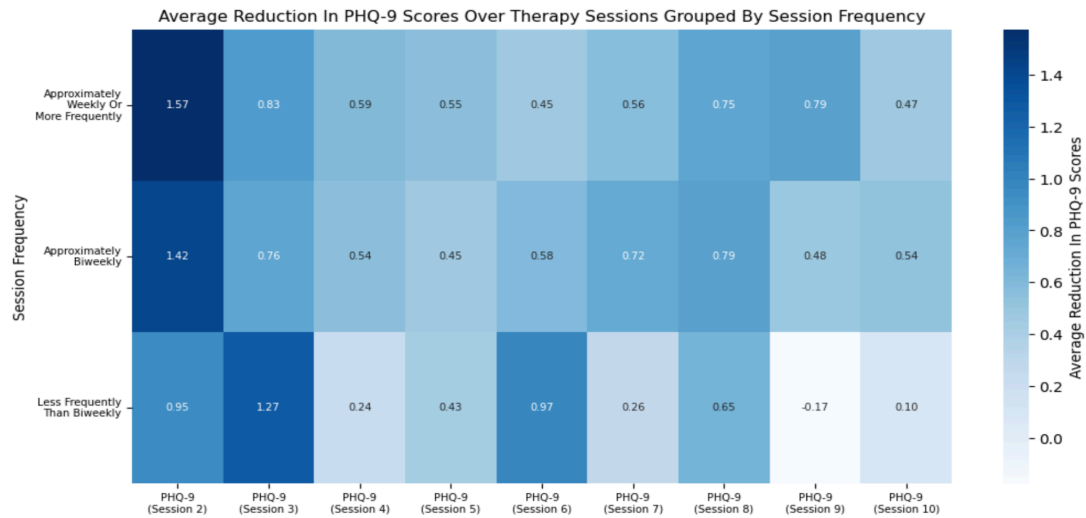
Note: Due to page limits, we will be focusing on the three most important visualizations that are most closely related to our problem statement and jobs to be done. For other visualizations that we have performed, refer to the GitHub link for the code, graphs and documentation: [Exploratory Data Analysis](#)

Visualization 1: Proportion Of Depression Among Age Groups In Singapore Over Time



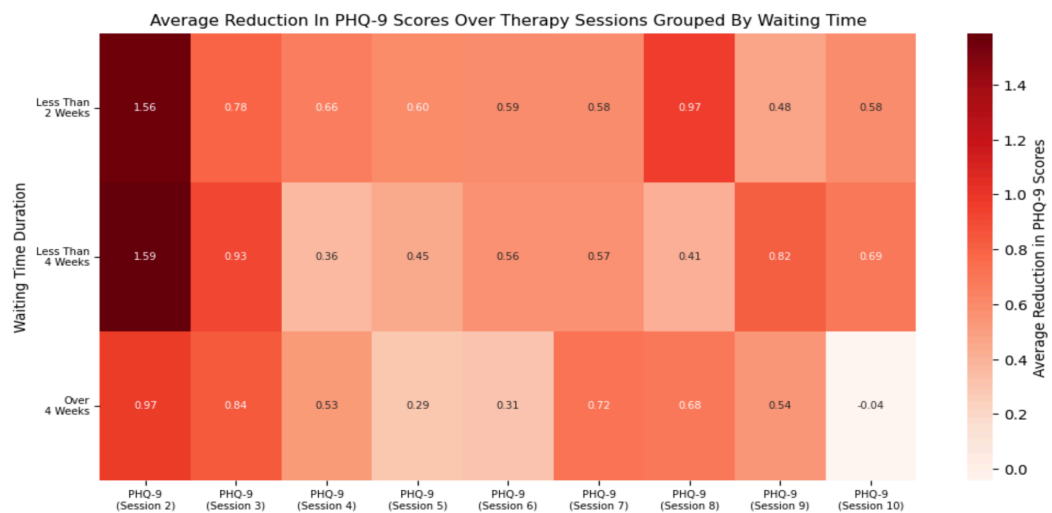
Analysis of Singapore's national depression prevalence data reveals a consistent and concerning pattern: the 15 to 24 age group carries the highest burden of depressive disorders across all measured years, peaking at 2.54% in 2021 before a marginal decline to 2.48% in 2023. This elevated prevalence is not incidental: it reflects the compounding effect of academic pressure, social transition and career uncertainty that characterises the university life stage, factors that are acutely present for NUS students. The spike observed between 2020 and 2021 further underscores the sensitivity of this demographic to external stressors, with the isolation, disruption and uncertainty of the COVID-19 pandemic producing a measurable deterioration in mental health outcomes that has not fully reversed. It is precisely this section of the population – NUS students in the 15 to 24 bracket – which is at the highest risk of depressive disorder, navigating an environment that actively compounds depression risk.

Visualization 2: Impact Of Session Frequency On Reduction In PHQ-9 Scores Across Sessions



Analysis of PHQ-9 score reductions across different counselling session frequencies reveals a clear and clinically meaningful relationship between session regularity and treatment outcomes. Students attending weekly or biweekly sessions demonstrate more consistent and sustained reductions in depressive symptom severity, with steady improvements maintained across the course of the programme. In contrast, students seen less frequently than biweekly exhibit highly irregular trajectories, including negative reduction values in later sessions indicating possible relapse, suggesting that extended gaps between appointments undermine the continuity of progress and leave patients vulnerable to deterioration going unaddressed. The largest PHQ-9 reductions occur immediately after the first session across all frequency groups, underscoring the critical importance of early and consistent engagement in preventing symptom escalation.

Visualization 3: Impact Of Waiting Time On Reduction In PHQ-9 Scores Across Sessions



Analysis of PHQ-9 score reductions across different pre-therapy waiting times reveals a compelling and clinically significant relationship between speed of access to support and treatment outcomes. Patients who began therapy within two weeks demonstrated the strongest and most consistent reductions in depressive symptom severity across the course of their programme while those who waited between two and four weeks showed moderate but less reliable improvements. Most critically, patients who waited longer than four weeks exhibited the weakest overall reductions and even negative values in later sessions, suggesting that prolonged delays before initial treatment not only slow recovery but may

actively contribute to symptom worsening and eventual relapse. The pattern is unambiguous: the sooner a deteriorating patient receives structured support, the better and more stable their clinical trajectory.

Linking Back To The Problem Statement And Jobs To Be Done:

Taken together, the three visualizations construct a compelling and evidence-based case for the intervention. The national prevalence data establishes that NUS students falling within the 15 to 24 age bracket represent the demographic at highest risk of depression in Singapore, confirming that this is the right population to target. The session frequency analysis demonstrates that students receiving less frequent counselling, mirroring the current UHC bi-weekly model, are the most vulnerable to irregular outcomes and relapse. The waiting time analysis then closes the argument: delays in receiving support are not merely inconvenient but clinically consequential, with longer gaps between a student's deterioration and their next point of care associated with measurably worse and less stable treatment trajectories. Collectively, these findings make clear that the core failure of the current UHC model – the absence of between-session monitoring and the inability to respond to deterioration in real time – is not a minor operational limitation, but a clinically significant structural gap.

2. Model Prototype

2.1 - System Overview

The proposed prototype directly addresses the structural gaps identified in the problem statement by introducing a between-session monitoring layer that operates alongside the existing UHC counselling programme rather than replacing it. Where the current model leaves students unmonitored across intervals of 7 to 14 days, creating conditions in which symptom deterioration goes undetected until the next scheduled appointment, the prototype intervenes by collecting a daily PHQ-2 signal from enrolled students, providing a continuous and clinically grounded touchpoint throughout the between-session period. This signal is processed through a rule-based triage engine augmented by an NLP chatbot, which classifies the student's current severity level and routes them to the appropriate intervention pathway in real time. The system tackles a failure of the existing model: the absence of any monitoring mechanism between sessions. The prototype is designed around three guiding principles: low friction for the student, proportionate response to severity and minimal additional burden on counsellors for stable cases, ensuring that limited clinical resources are directed toward students who need them most urgently while preserving the accessibility of the support experience for all enrolled users.

2.2 - Signal: PHQ-2 Daily Check-In

The PHQ-2 Daily Check-In serves as the primary screening signal, where students report daily on two validated questions each scored from 0 ("not at all") to 3 ("nearly every day"), producing a composite score from 0 to 6. The questions are as follows:

- 1) Over the past few days, how often have you been bothered by feeling down, depressed, or hopeless?
- 2) Over the past few days, how often have you been bothered by little interest or pleasure in doing things?

Scores of 0 to 2 are classified as "Well", requiring no immediate clinical intervention. The student is automatically provided with self-care resources including psychoeducation articles, mood-boosting activities and mindfulness exercises, with no counsellor involvement required.

For scores of 3 to 6, rather than treating the PHQ-2 score as a direct escalation trigger, the system routes all such students into a dynamic chatbot triage interaction. This design decision is deliberate and addresses a significant vulnerability of purely score-based systems: the risk of gaming, where a student selects the highest possible responses on the PHQ-2 specifically to trigger faster access to a counsellor, irrespective of their actual clinical state. Under a conventional cutoff-based approach, a student who answers "nearly every day" to both questions would be automatically escalated regardless of whether their responses reflect genuine clinical severity. By extending the chatbot triage layer across the entire 3

to 6 range, the system ensures that escalation decisions are grounded in the richer, harder-to-fabricate signal of natural language rather than two survey items alone. Within the chatbot interaction, the student engages in a dynamic conversation for a fixed number of turns. Questions are generated and sequenced based on the student's preceding responses, probing clinically relevant dimensions such as mood trajectory, functional impairment, sleep disruption, anhedonia and the presence of hopelessness or passive suicidal ideation.

The concatenation of the student's responses is then processed by an NLP classifier trained on the Reddit and Twitter Depression Severity Levels Dataset, which produces a severity score on a four-point scale of 0 to 3. This secondary score, rather than the raw PHQ-2 score, determines which intervention pathway the student is routed into as described in the Signal-to-Decision Table below. This layered approach preserves clinical safety by ensuring that all students scoring 3 or above are assessed further, while significantly improving the specificity of escalation decisions by introducing a natural language signal that is substantially more difficult to game than a structured survey.

2.3 - Signal-to-Decision Table

Tier	PHQ-2 Range	Chatbot NLP Score	Description Of Outcome
1 (Well)	0 to 2	-	Provide self-care resources
2 (Grey Zone → Stable)	3 to 6	0	- Chatbot triage activated - NLP classifier determines that score = 0 - Provide self-care resources - Provide emergency hotlines
3 (Grey Zone → Mild)	3 to 6	1	- Chatbot triage activated - NLP classifier determines that score = 1 - Provide option to book appointment with counsellor - Students without an appointment can book a new one - Student with an upcoming appointment can push it forward - Provide self-care resources - Provide emergency hotlines
4 (Grey Zone → Moderate)	3 to 6	2	- Chatbot triage activated - NLP classifier determines that score = 2 or 3 - Provide option to speak to counsellor, via live chat - Provide option to book appointment with counsellor (see Tier 3) - Provide self-care resources - Provide emergency hotlines
5 (Grey Zone → Critical)	3 to 6	3	

2.4 - NLP Model for Grey Zone Triage

Should the chatbot triage be triggered, we shall proceed as follows. Upon concatenating the student's responses from the chatbot conversation, we will process this textual input using a final-stage NLP model. The resultant output yields a secondary score that describes the severity of the student's depression status. This prediction will inform the subsequent actions that are to be taken by the application, ensuring that our interventions assist the user in a more targeted manner. To design a robust model for this feature, we explored two approaches.

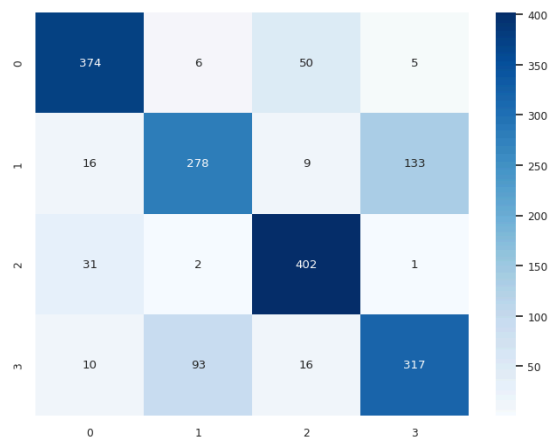
Fine-Tuning BERT

Parameter-efficient fine-tuning (PEFT) was conducted on the `google-bert/bert-base-cased` encoder architecture. We made use of the Reddit and Twitter Depression Severity Levels Dataset, which contained a total of 41859 textual samples, each being associated with a depression severity label (0-3). Therefore, our goal is to have the NLP model learn to predict the correct level of severity, for a given textual input.

There are two ways to carry out the fine-tuning process. Firstly, we can attach a classification head to the end of the BERT encoder - in this case, the task reduces to a 4-class classification problem, and cross entropy loss is used as the criterion. Alternatively, since severity labels are ordinal, we can make use of a regression head instead, so as to preserve some notion of ordering. For this, mean squared error will be our criterion of choice. During inference, the regression output from this model will be rounded off, so as to generate the prediction required. On this note, we explored both of these options. For both trials, Low-Rank Approximation (LoRA) was employed, speeding up the training process by freezing a significant proportion of model parameters. Both models were fine-tuned for 10 epochs, with a learning rate of 1×10^{-4} .

Trial	Method	Accuracy	Precision	Recall	F1
1	Classification	<u>0.787</u>	<u>0.785</u>	<u>0.787</u>	<u>0.784</u>
2	Regression	0.564	0.654	0.564	0.561

We see that the fine-tuned classifier outperforms the regression model across all metrics. Thus, the fine-tuned classifier was selected as our model of choice.



Confusion Matrix of Fine-Tuned Classifier

With respect to the model selected, Class 2 performs the best (recall 92.2%) and Class 1 performs the worst (recall 63.8%). The three most important error patterns are shown below.

- **Class 3 downgraded to Class 1 (93 cases – the most clinically dangerous errors).** High-severity posts are missed when their language is restrained or conditional. For example, “I truly have never been more depressed in my life, but I cannot bring myself to kill myself because I have to take care of my cats” had been annotated as Class 3. The speaker actively expressed clear suicidal intent, even if they were hesitant on following through with it. However, the model incorrectly assigned Class 1 to this textual input. Evidently, the classifier appears to underweight suicidal content when it is implicitly hedged or embedded in a longer emotional narrative.

- **Class 1 over-escalated to Class 3 (133 cases).** The model frequently misclassifies moderate depression as severe when the text contains explicit death-related or method-related language, even without genuine suicidal intent. For instance, consider “Like would it be asphyxiation and hemorrhage and airway obstruction?... Theoretically, what is the most foolproof way to die?”, which had been annotated as Class 1. From an objective standpoint, this post merely posed a hypothetical and intellectual question. However, it was predicted as Class 3 – this could be attributed to the fact that it discussed the concept of self-harm in an overly blunt manner. The model latches onto method-specific vocabulary as a severity signal, accounting for intent to only a limited extent.
- **Class 0 misclassified as Class 2 (50 cases).** Some neutral posts are falsely flagged, likely due to insufficient text for the model to classify confidently. For instance, “Tfw you meet one of your favorite random twitter follows”, a short and neutral post was predicted as Class 2. With little lexical signal to work with, the model appears to default toward a mid-range depression class rather than Class 0.

SentenceBERT Embeddings

We also attempted to use another model for the Grey Zone Triage. First, from the test dataset, all the text samples are fed into SentenceBERT (SBERT) in order to obtain individual embeddings. Dimension reduction is then done on the embeddings, and the result embedding is then used to form a logistic regression model. Its accuracy on validation data (0.253) is rather low.

Validation Accuracy: 0.2525832376578645				
	precision	recall	f1-score	support
0	0.43	0.13	0.20	436
1	0.25	0.28	0.26	435
2	0.24	0.50	0.32	436
3	0.22	0.10	0.14	435
accuracy			0.25	1742
macro avg	0.28	0.25	0.23	1742
weighted avg	0.28	0.25	0.23	1742

Therefore, GridSearchCV is used to find better hyperparameters which will get the model to have higher validation accuracy. As a result, the final model has managed to achieve a much higher validation accuracy of 0.685 as shown in figure below titled “GridSearchCV Result”.

```
Best params: {'clf__C': 0.01, 'pca__n_components': 350}
Validation score: 0.6847697555002644
```

GridSearchCV Result

With hyperparameters that yield the greatest validation score, we use the model with these hyperparameters and manage to achieve a relatively high accuracy on the testing set (0.683), as shown below.

Accuracy: 0.6833046471600689

Classification Report:				
	precision	recall	f1-score	
0	0.77	0.82	0.79	
1	0.57	0.57	0.57	
2	0.75	0.73	0.74	
3	0.64	0.61	0.62	
accuracy			0.68	
macro avg	0.68	0.68	0.68	
weighted avg	0.68	0.68	0.68	

Even with improved test accuracy, the fine-tuned BERT classifier still has the higher test accuracy. Thus, the fine-tuned BERT classifier is chosen as our model.

2.5 - Success Metrics

Metric 1: Reduction in Response Delay for High-Severity and Deteriorating Students

Objective: Measure whether the system meaningfully reduces the time between symptom deterioration and appropriate clinical response specifically for students whose condition is worsening, directly addressing the monitoring blind spot identified in the problem statement.

Computation: For each student flagged as Tier 4 or Tier 5 in the signal-to-decision table (NLP severity score of 2 or 3), calculate the time elapsed between the check-in response that triggered escalation and the delivery of a counsellor intervention or priority appointment offer. In the pre-rollout period, the equivalent delay is approximated by identifying students in the CBT baseline dataset whose PHQ-9 scores worsened between two consecutive sessions and calculating the remaining interval between the inferred point of deterioration and the next scheduled session, based on UHC's standard bi-weekly scheduling of 7 to 14 days. Since deterioration can occur at any point during the between-session interval, this approximated delay reflects the structural ceiling imposed by fixed-interval scheduling in the absence of any monitoring mechanism.

Comparison: Compare the mean response delay for high-severity and deteriorating students pre-rollout against post-rollout. **Given that counsellor hours are fixed**, this metric also tracks whether prioritisation of Tier 4 and Tier 5 students results in a reallocation of counsellor availability rather than an overall increase, ensuring that reduced delay for high-risk students is not simply achieved by displacing stable students. To verify this, response delay for Tier 1 students is monitored in parallel: if Tier 1 delays remain broadly stable or increase only marginally, it confirms that the system is reallocating existing capacity toward students who need it most urgently rather than requiring additional counsellor resources which would not be operationally realistic.

Success Criterion: Post-rollout average response delay for Tier 4 and Tier 5 students is significantly shorter and more consistent than the pre-rollout approximated delay of 7 to 14 days, ideally reducing to same-day or within 1 to 2 days. Tier 1 response times remain broadly stable, confirming that improved responsiveness for deteriorating students is achieved through reallocation rather than resource expansion. This directly validates JTBD 1: that structured between-session check-ins enable the detection of early warning signs and the delivery of timely intervention before the student's condition escalates to a crisis point.

Metric 2: PHQ Improvement Relative to Baseline, Controlling for Natural Recovery

Objective: Determine whether between-session monitoring and early escalation produces measurably greater clinical improvement in depression severity compared to the standard care pathway, whilst

accounting for the well-documented tendency of PHQ scores to improve over time even without structured intervention.

Computation: Group post-rollout students by initial severity band using their first recorded PHQ-9 score: mild (5 to 9), moderate (10 to 14) and severe (15 to 19). This stratification is necessary because students with higher starting severity have greater room for improvement and exhibit different recovery trajectories compared to those with milder presentations; aggregating across severity bands would obscure these differences and produce misleading results. For each band, calculate the average PHQ delta (first session score minus last session score) at equivalent programme timepoints, controlling for the number of sessions attended and total programme duration to ensure comparability. A positive delta indicates improvement in depressive symptom severity. To account for natural recovery as a confounder, compute the same PHQ delta for a matched baseline cohort drawn from the CBT dataset: students of the same initial severity band who completed a comparable number of sessions under the standard fixed-interval model without access to the check-in system.

Comparison: Compare the average PHQ delta of post-rollout students against the matched baseline cohort within each severity band. The baseline cohort controls for the natural trajectory of PHQ improvement under standard care, isolating the incremental clinical benefit attributable to early detection and escalation. The difference between the two deltas within each band represents the improvement that can be attributed to the intervention rather than to natural recovery or the effects of counselling alone.

Success Criterion: Post-rollout students in the moderate and severe bands show a meaningfully greater average PHQ improvement than matched baseline students of the same starting severity and equivalent session count, with the most pronounced improvement expected in the higher severity bands. This is because students with moderate to severe depression are the most vulnerable to the consequences of the current 7 to 14 day monitoring gap: they are the ones most likely to deteriorate significantly between sessions, and therefore stand to benefit the most from early detection and timely escalation. If students in the mild band show a smaller or negligible difference compared to baseline, this is expected and consistent with the design intent, as milder presentations tend to improve substantially under standard care and the marginal benefit of accelerated intervention is inherently smaller for this group. This directly validates JTBD 2: that students in a particularly difficult period receive care proportionate to their severity, and that this proportionate, timely care translates into measurably better clinical outcomes than the one-size-fits-all fixed-interval scheduling model.

3. Product Interface

3.1 - Overview Of Product Interface

The prototype is designed as a **mobile application** serving two distinct user groups: NUS students enrolled in UHC counselling for depression and UHC counsellors managing their caseload remotely. The interface is clean and structured, with a calm visual language appropriate for a mental health context. We constructed the application manually in Figma. ([Link to Figma Interface](#))

Student-Facing Interfaces:

The student-facing side includes a **daily PHQ-2 check-in flow**: a brief, low-friction prompt that captures the student's current mood and depression symptom severity. A chatbot interface serves as the primary conversational layer, activated for all PHQ-2 scores of 3 to 6, guiding students through a dynamic adaptive conversation and producing a secondary NLP severity score of 0 to 3 that determines the intervention pathway assigned to the student., offering either reassurance and self-help resources for stable presentations or escalation prompts for worsening ones.

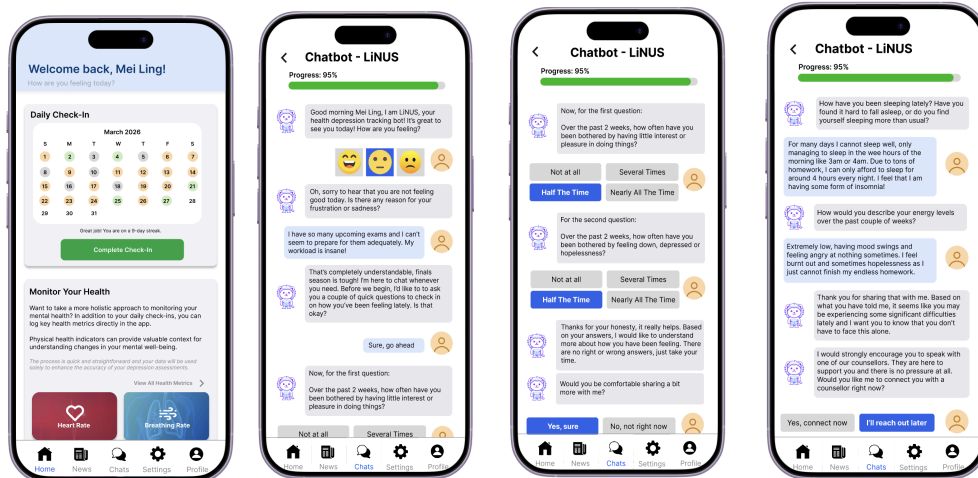


Image 1: Home page of the NUS Depression App where users complete their daily check-in (brings to the chatbot page)
 Image 2: User answering some initial questions by the chatbot to make them feel comfortable before the actual PHQ-2 questions are given
 Image 3: PHQ-2 questions administered by the chatbot where the user selects an option for each question for the chatbot to assess
 Image 4: User answering free-text questions to the chatbot for those who are classified under the "grey zone pathway"

For stable cases, the chatbot offers reassurance and signposts relevant self-help resources such as bringing users to the news page where they can find resources pertaining to mental health and depression. For worsening cases, the chatbot delivers escalation prompts proportionate to the detected severity level. In instances where the chatbot identifies severe depressive symptoms, the system activates an elevated response pathway: the student is immediately presented with the option to connect with an available UHC counsellor for real-time support if they wish to do so, alongside direct access to mental health helpline numbers for immediate external assistance. Recognising that not all students in crisis will feel ready for immediate contact, the interface also provides the flexibility to book an upcoming counselling appointment at a preferred date and time, lowering the barrier to accessing structured support without imposing it. This layered response design ensures that students at the highest level of need are never left without a clear and accessible next step while preserving their autonomy in choosing the form of support that feels most appropriate to them at the moment.

At the same time, the system design also takes into account the operational realities on the counsellor's end. Rather than assuming that all booked appointments can be accommodated, the platform can incorporate a validation mechanism where only pre-verified available time slots are open for booking. In this approach, a counsellor's availability is reflected in real time, with unavailable slots greyed out to prevent scheduling conflicts. This reduces the need for manual confirmation while maintaining reliability in the booking process.

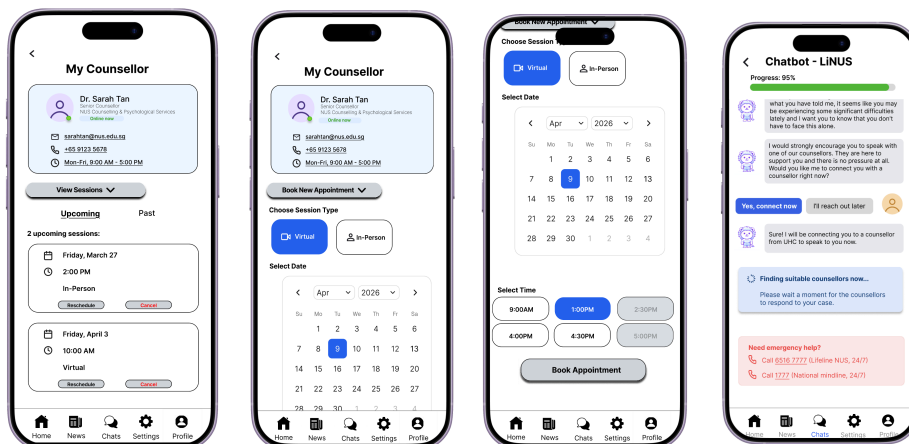


Image 1: Booking a counselling session (user has the freedom to choose date and time)
 Image 2: User booking/rescheduling an appointment with the counsellor once prompted
 Image 3: User only able to select available time slots given with reference to counsellor's schedule
 Image 4: Helplines of Lifeline NUS and National Mindline given, chatbot finding a suitable counsellor to a user who prefers immediate counselling

Moreover, the **news and resources** section functions as a curated mental health knowledge feed, surfacing articles and updates from reputable and clinically credible sources such as the Institute of Mental Health (IMH), the World Health Organisation (WHO) and other trusted public health bodies. Content is focused on mental health awareness, the nature and treatment of depression and evidence-based strategies for managing depressive symptoms. Engagement with this section is entirely optional, it is designed for students who wish to take a more active role in understanding their condition, equipping them with reliable, accessible knowledge rather than leaving them to navigate unverified sources independently.

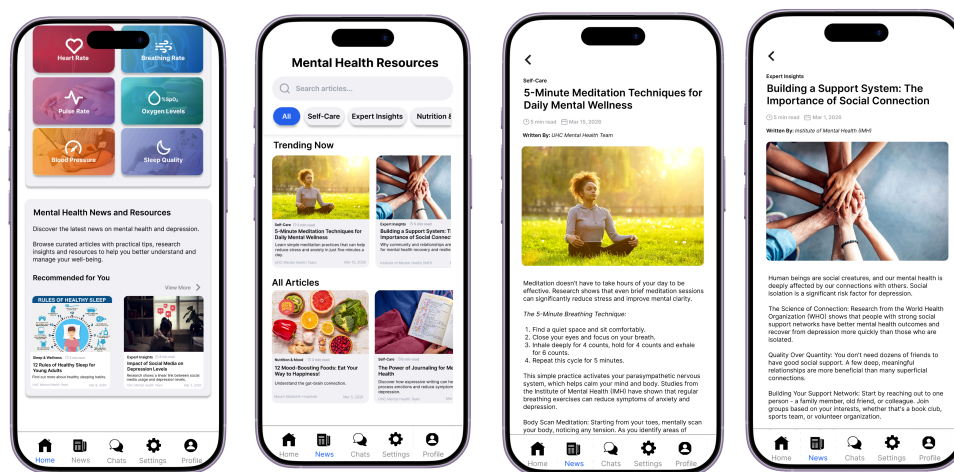


Image 1: Mental Health News And Resources section in the home page of the app
 Image 2: The main mental health resources page, where users can browse news articles by topic (self-care, expert insights etc.)
 Image 3: A specific health article by the UHC mental health team about meditation techniques
 Image 4: Another specific health article by the Institute of Mental Health (IMH) regarding the importance of social connections

Counsellor-Facing Interfaces:

The counsellor-facing portal provides a structured overview of their active student caseload, with triage indicators clearly flagging students who have been escalated based on recent PHQ-2 check-in data, enabling counsellors to prioritise their limited availability toward those with the most pressing clinical need. A **shared chat and messaging interface** bridges the communication gap between scheduled sessions, allowing students to reach out directly to their assigned counsellor when they feel the need for support or guidance outside of formal appointments. This asynchronous messaging capability ensures that students are never left without a point of contact during the between-session period. In higher-acuity situations, specifically where a student's PHQ-2 check-in has flagged severe depressive symptoms and the student has expressed willingness to speak to someone immediately, the platform supports the initiation of a real-time online counselling session, enabling the counsellor to deliver timely, direct support without requiring the student to wait for a scheduled in-person appointment. This dual-mode communication model – combining asynchronous messaging for general support with on-demand live sessions for crisis-level presentations – allows counsellors to respond proportionately to the full spectrum of student's needs, whilst maintaining a clear overview of their caseload at all times.

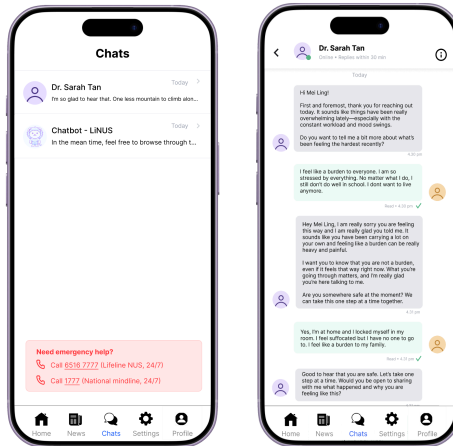


Image 1: The chat section of the app where users can interact with their counsellors or the chatbot directly

Image 2: An in-depth chat of the user with his/her assigned counsellor Dr Sarah Tan, users can chat using the app directly

3.2 - Human-in-the-Loop Design

At what tiers does the system act autonomously versus flag for human review?

The system is designed to operate autonomously for stable, low-severity scenarios, specifically Tier 1 cases where PHQ-2 check-in responses indicate no significant deterioration - delivering reassurance messages, self-help resources and curated mental health content without requiring any counsellor involvement. For students whose NLP severity score is 0 to 1 (Stable or Mild), the system responds autonomously: delivering self-care resources and surfacing appointment booking options without direct counsellor action unless the student initiates contact. It is at Tier 4 and Tier 5, where the NLP severity score reaches 2 (Moderate) or 3 (Critical) that the system transitions from autonomous operation to active human flagging. At this tier, the counsellor's dashboard is updated in real time with an escalation alert, the student is offered immediate counsellor contact or helpline access and the case is surfaced as a priority within the counsellor's caseload view, requiring human acknowledgement and response.

How does the counsellor confirm, override or escalate a system-generated alert?

Upon receiving an escalation alert on their dashboard, the counsellor reviews the flagged student's recent check-in data, PHQ score trajectory and any chat messages sent by the student. To enhance transparency, the counsellor can also access the prediction probabilities from the NLP classifier's output, enabling them to view the outcome of the model's decision-making process. With this information, the counsellor has the autonomy to form a clinical judgement about the appropriate level of response. The counsellor can confirm the alert by initiating a real-time online counselling session through the platform, or by sending an immediate message to the student acknowledging the flag and arranging follow-up. Where the counsellor's clinical assessment differs from the system-generated triage outcome – for instance, if contextual knowledge of the student's history suggests the flag may not reflect a genuine crisis – the counsellor retains full authority to override the escalation and reclassify the case, adjusting the student's triage tier accordingly. Conversely, if the counsellor determines that the situation is more severe than the system has captured, they can manually escalate the case further by arranging an urgent in-person appointment or coordinating with UHC clinical staff for a higher level of intervention. All counsellor actions: confirmations, overrides and escalations are logged within the system to maintain a transparent and auditable record of clinical decision-making.

What prevents the system from acting on a false positive without counsellor awareness?

The system incorporates several safeguards to ensure that no high-stakes automated action is taken on the basis of a potentially erroneous triage output, without counsellor awareness. For escalations, the system does not autonomously initiate contact with external services or make clinical decisions on the

student's behalf. It surfaces the alert to the counsellor's dashboard and presents the student with options, but the decision to engage in a live counselling session or to act on the escalation remains with both the student and the counsellor. For users who are initially classified under the grey zone, the dual-signal design – combining the rule-based PHQ-2 triage engine with the NLP chatbot's independent linguistic and emotional assessment – lowers the possibility of having a false positive, as both signals are considered in determining severity. Furthermore, because the chat interface allows students to communicate directly with their counsellor at any time, a counsellor who receives an escalation alert can immediately contextualise it against the student's recent messages and known history before deciding how to respond, rather than acting solely on the system's output.

4. Safety and Ethics

4.1 - Intended Use And Scope Of Model & App

Who is this system designed for?

In its current pilot phase, the system is designed exclusively for NUS students enrolled in UHC counselling for depression and the UHC counsellors managing their care. This deliberately constrained scope reflects the need to validate both the clinical effectiveness of the triage model and the usability of the application within a controlled, manageable cohort before broader deployment. A smaller initial sample is critical: it allows the team to identify model limitations, surface usability issues and assess clinical outcomes without the risks associated with premature large-scale rollout. Continuous monitoring of system performance, triage accuracy and student outcomes will be conducted throughout this phase to inform evidence-based decisions about progression. If the pilot demonstrates sufficient effectiveness, the system will be expanded in structured stages: Phase 2 would extend access to university students across all Singapore universities (NTU, SMU, SIT and others), in partnership with their respective campus health services. Phase 3 would broaden the scope further to encompass all Singapore youths aged 15 to 24, including those outside the university system, reflecting the epidemiological evidence that this age band represents one of the highest-prevalence demographic segments for depression in Singapore. Rigorous and continuous monitoring remains a non-negotiable requirement at every phase of rollout, ensuring that the system's clinical impact is consistently evaluated and that any deterioration in model performance or user outcomes is detected and addressed promptly.

What this system is explicitly not?

The system is not a replacement for professional therapy and it is not designed to function as one. Counsellors remain central to the care pathway at every stage: the system's role is to support and augment their clinical judgement, not to supplant it. All significant decisions, including the confirmation of escalation alerts, the initiation of counselling sessions and the interpretation of triage outputs remain firmly within the counsellor's authority. The system is also not intended to function as a standalone diagnostic tool. While the daily PHQ-2 check-in and NLP chatbot layer can surface clinically meaningful signals and help counsellors identify deteriorating cases more efficiently, these outputs are designed to inform rather than replace formal clinical assessment. Importantly, the system addresses a number of practical limitations of the current model: by enabling asynchronous communication and online counselling sessions, it reduces the dependency on physical appointments, accommodating students who may prefer or require remote interaction. This also serves to reduce pressure on the UHC system by decreasing the volume of in-person sessions needed for stable or mildly deteriorating cases, allowing counsellor capacity to be concentrated where it is most clinically necessary without overwhelming existing infrastructure.

How is access granted and managed?

Access to the system is restricted to students who are currently enrolled in UHC's counselling programme for depression. It is granted on an opt-in basis following enrolment through UHC's standard intake

process. Students are not able to self-register independently of the clinical pathway, ensuring that all users are already under the care of a UHC counsellor and that the system operates as an extension of an existing therapeutic relationship rather than a standalone service. Counsellor accounts are provisioned directly by UHC with access tied to their active caseload. This access control model ensures that the system remains within the bounds of a supervised clinical framework at all times, preserving both the safety of students and the integrity of the data collected.

4.2 - Handling High-Risk and Outlier Cases

What happens if a student receives a high NLP severity score repeatedly over multiple days?

Under the updated framework, a PHQ-2 score of 5 to 6 alone does not trigger automatic escalation: all scores in the 3 to 6 range enter the chatbot triage layer and it is the NLP severity score that determines the intervention. However, a sustained pattern of high NLP severity scores (2 or 3) across multiple consecutive days is treated as a significant clinical signal warranting human review. When such a pattern is detected, the system automatically flags the case to the counsellor's dashboard, highlighting both the repeated high NLP scores and the duration over which they have persisted. The counsellor is prompted to intervene directly, reviewing the student's full check-in and chatbot history and where appropriate initiating a direct message through the platform to enquire about the student's current state. This human verification step is deliberately built into the workflow, acknowledging that even within the chatbot layer, sustained high-severity language patterns require clinical judgement to interpret rather than automated action alone.

What if a student does not respond to check-ins for several days?

A prolonged absence of check-in responses is treated as a potential indicator of silent deterioration, a scenario in which a student's condition may have worsened to the point where engagement with the app feels inaccessible or overwhelming. The system's first response is a gentle, non-intrusive in-app prompt encouraging the student to complete their daily check-in. This is accompanied by a message reassuring them that their responses help UHC better understand and support their mental health and that there is no pressure to provide a particular answer. The tone of this prompt is deliberately supportive rather than procedural, recognising that a student who is struggling may be particularly sensitive to feeling surveilled or judged. If the student remains unresponsive after a defined number of days, the system escalates the non-response to the counsellor's dashboard as a potential silent deterioration flag. The counsellor is then prompted to reach out directly via the platform's messaging interface: not to demand a check-in response, but to check in on the student on a personal level – asking simply whether they are doing alright and whether there is anything they need. The student can respond at their own pace and the counsellor uses the content and tone of that exchange to determine whether the situation warrants a more formal clinical intervention, an expedited appointment or continued gentle monitoring.

How does the chatbot NLP model reduce misclassification of a worsening case as stable?

Misclassification of a worsening case as stable is the most clinically consequential failure mode of the system and the architecture incorporates multiple overlapping safeguards to reduce both its likelihood and its impact. The risk of NLP misclassification applies across the full chatbot triage range (PHQ-2 score 3 to 6), since the NLP severity score is now the primary determinant of intervention for all students in this range. Within this grey zone, the NLP model poses a fixed number of free-text questions, deliberately kept concise to avoid overburdening or distressing a student who may already be in a vulnerable state. If the model reaches a confident classification within this sequence, producing an NLP severity score of 0 or 1 (in which case no counsellor alert is raised) or a score of 2 or 3 (in which case an escalation flag is triggered), the interaction concludes accordingly. However, the model might instead fail to confidently assign a severity classification – when the probability of the predicted class is below a threshold, for example. In this exceptional case, the system defaults to flagging the student as potentially depressed, escalating the case for counsellor review.

This design reflects a deliberate and clinically grounded asymmetry in how the system treats uncertainty: a false positive (incorrectly flagging a student who is not in crisis) results in an unnecessary but manageable counsellor touchpoint, causing the student minor inconvenience or momentary worry. A false negative (failing to flag a student who is genuinely deteriorating) risks allowing a worsening depressive episode to go undetected and unaddressed with potentially life-threatening consequences. When the cost of the two error types is this unequal, erring on the side of caution is not merely a technical preference but a clinical and ethical imperative. Despite being less of a concern, additional measures are still being taken to address model hallucinations associated with false positives. With reference to Section 3.2 (Human-In-The-Loop Design), the counsellor always has the final say in determining the severity of a given user's depression, upon review. The addition of a last-stage "human-in-the-loop" guardrail not only enhances transparency, but also mitigates the effects of false positives resulting from model hallucinations.

4.3 - Use of Red Teaming

Having conducted a round of error analysis for the NLP classifier in Section 2.4 (NLP Model for Grey Zone Triage), we now have a better understanding of our solution's potential pitfalls. With respect to these exceptional cases, we aim to construct red team scenarios to stress-test the system's decision-making process. We shall make use of inputs that are more likely to confuse the model – examples include:

- Inputs that use death-related language, but merely in a hypothetical or academic framing
- Inputs where suicidal intent is present, but implicitly hidden in longer narratives

Moreover, we can simulate adversarial or unreliable user behaviour in the upstream PHQ-2 check-in phase. This can be achieved by consistently selecting maximum scores regardless of true clinical state. In this manner, the red teaming exercise will further expose weaknesses in our pipeline. These insights can inform subsequent refinements and guide the collection of additional training data for underperforming scenarios, thereby improving the robustness of our intervention.

4.4 - Data Privacy and Consent

What data is collected, stored, and for how long?

The system collects two primary categories of data from enrolled students: PHQ-2 daily check-in responses (including any longitudinal PHQ score trajectories generated over the course of the student's enrolment) and free-text inputs submitted through the chatbot interface during check-in interactions. Supplementary health indicators logged voluntarily through the health tracking dashboard such as sleeping hours, heart rate and breathing rate are also retained where the student chooses to provide them. All collected data is stored securely within UHC's internal systems and is retained only for as long as the student remains actively enrolled in the counselling programme and requires ongoing clinical support. The retention period is therefore not fixed by a predetermined timeframe but is instead tied directly to the student's continued engagement with UHC, reflecting the clinical rationale that longitudinal data is necessary to track the student's progress and inform treatment decisions over time. Once a student is assessed as no longer requiring UHC support and is formally discharged from the programme, all associated data is permanently and securely erased from the system in accordance with proper data disposal protocols, minimising the risk of residual exposure of sensitive healthcare information.

Who has access to individual student data?

Access to a student's data is strictly limited to the UHC counsellor directly responsible for that student's care. No other counsellors, UHC administrative staff or third parties outside of UHC are granted access to an individual student's PHQ scores, chatbot responses or health tracking inputs. This principle of minimum necessary access is applied rigorously given the highly sensitive nature of mental health data where unauthorised disclosure carries significant risk of harm to the student. Data is siloed at the

individual counsellor level within the system, ensuring that even within UHC, a student's information remains visible only to those with a direct and active clinical relationship with that student.

How is student consent obtained and can it be withdrawn?

Consent is obtained through an explicit, in-person conversation during the student's initial onboarding session with their UHC counsellor prior to any data collection taking place. The counsellor clearly explains what data will be collected, how it will be used, who will have access to it and how long it will be retained. The counsellor will directly ask the student whether they consent to this collection. Consent is entirely voluntary: students who do not wish for their data to be collected are under no obligation to agree and UHC is not permitted to collect or process any student data without prior explicit consent. Students retain the right to withdraw their consent at any point during their enrolment, at which point data collection ceases immediately and any previously collected data is handled in accordance with the same secure disposal protocols applied.

NOTE: The system is designed and operated in full compliance with Singapore's Personal Data Protection Act (PDPA), which governs the collection, use, disclosure and storage of personal data. Given that the data handled by the system constitutes sensitive healthcare information, the highest standards of data protection are applied throughout, including secure storage, strict access controls, explicit consent requirements and defined data disposal procedures. The system additionally adheres to NUS's internal data governance policies which set out the institution's obligations with respect to the handling of student data.

4.5 - Potential Biases In Model Used

1. Training Data and Platform Mismatch (Most Critical)

The NLP classifier is trained on text posts sourced from Reddit and Twitter: platforms characterised by informal, often anonymous and uninhibited self-expression. Users on these platforms frequently employ colloquialisms, slang, abbreviations and emotionally unfiltered language when discussing their mental health. However, the chatbot interface within this application operates in a fundamentally different communicative context: students are aware that their responses may be reviewed by a professional UHC counsellor which is likely to produce more measured, formal and self-edited language. This platform affordance mismatch means the model has been trained predominantly on an expressive register that may not reflect how NUS students actually communicate within the app, potentially reducing the classifier's sensitivity to distress signals that are conveyed in more restrained or formal language. For NUS's diverse student population, which encompasses students from varied cultural backgrounds, linguistic traditions and comfort levels with emotional disclosure, this mismatch is likely to manifest unevenly. Students who express themselves in ways that diverge most significantly from the Reddit and Twitter training distribution are at greater risk of being misclassified, particularly in false negative cases where genuine distress is expressed in ways the model has not been adequately trained to recognise.

2. Cultural & Linguistic Bias

A significant limitation of the model lies in its lack of alignment with the cultural and linguistic context of the NUS student population. Since the training data is largely drawn from Reddit and Twitter, platforms that are predominantly Western-centric, the model may not adequately capture the nuances of local expression in Singapore. NUS students often communicate using a blend of standard English, Singlish and culturally specific phrases ("sian", "damn shag", "cannot take it already") which may carry emotional weight that differs from their literal meaning. Additionally, cultural norms in many Asian contexts tend to favour indirect or restrained expressions of distress rather than explicit statements of negative emotions. As a result, the model may fail to recognize subtle indicators of mental health risk or misinterpret culturally embedded language, leading to false negatives or inaccurate classifications.

This bias can disproportionately affect certain student groups, thereby reducing the overall reliability of the classifier across NUS's diverse population.

4.5 - Ethical Considerations

A key ethical concern is the risk of **over-reliance on automated triage systems**. If students begin to place excessive trust in the NLP classifier's outputs, they may perceive it as a sufficient substitute for professional judgement and delay or avoid seeking help from human counsellors. This is particularly problematic in mental health contexts where nuanced understanding, empathy and contextual judgement are critical. Over-reliance could also lead to a false sense of security if the model produces false negatives, causing at-risk students to believe their condition is not serious. In a diverse environment like NUS, where help-seeking behaviours already vary across cultural groups, such reliance may further discourage vulnerable students from engaging with formal support systems, undermining the intended purpose of the intervention.

Another important ethical dimension is the **psychological impact of classification labels**, particularly when a student is flagged as "Critical". While intended to trigger timely intervention, such labels may inadvertently increase anxiety, distress or self-stigma. Being categorised in a high-risk group could make students feel overwhelmed, judged or even hopeless, especially if the classification is presented without adequate context or immediate human support. There is also a risk of misclassification, where incorrect labelling may either alarm students unnecessarily or fail to validate their experiences. Therefore, careful consideration must be given to how results are communicated: framing outputs in a supportive, non-deterministic manner and ensuring that high-risk classifications are always accompanied by clear guidance, reassurance and access to human counsellors.

5. Learning and Reflections

5.1 - Key Learnings

Over the course of the project, our team developed a deeper understanding of both the problem and the users we are designing it for. One of the surprising insights we had was that not all students experiencing depression are inclined to seek help, even when resources are readily available. This challenged our initial assumption that accessibility alone is sufficient to drive help-seeking behaviour. However, we also recognised that the decision to seek help ultimately lies with the user's own autonomy. While our application functions as an intervention, it cannot override individual willingness to engage with support.

Initially, we focused on an all-in-one application that covers all aspects of health, which resulted in our solution being overly broad and insufficiently tailored to the target audience. As the project progressed, we realised the importance of focusing more deliberately on the specific pain points of students in distress and refining our scope accordingly. In particular, rather than relying solely on static screening tools such as the PHQ-2 to triage users, we introduced a layer of dynamic questioning for cases that fall into ambiguous ranges. This allows the system to better differentiate between varying levels of need, improving the accuracy of triaging outcomes. At the same time, this approach helps mitigate potential misuse or exploitation of the system, ensuring that high-priority resources are directed appropriately. Given that counsellor availability is inherently limited, our design prioritises access to live chat and appointment booking for users who are assessed to be at higher levels of risk. This ensures that those in more severe situations are triaged efficiently to professional support, while still providing alternative forms of assistance for users with lower or less urgent needs.

5.2 - Limitations of the Current Prototype

A key limitation lies in the **use of the PHQ-2 as the primary screening tool** as it is designed for quick preliminary assessment rather than diagnosis. With only two questions, it may not comprehensively capture the current mental health status of the user. While the system attempts to mitigate this limitation by incorporating free-text chatbot interactions for all PHQ-2 scores of 3 to 6, this approach is still

imperfect. Many important factors influencing mental health – such as genetic predisposition, family history, environmental stressors and personal circumstances – are not captured by either the PHQ-2 or short conversational inputs. As a result, the triage signal may lack reliability, leading to false positives (causing unnecessary anxiety or alarm) or false negatives (failing to identify at-risk individuals). In the latter case, students may remain untreated if the system incorrectly deems them low-risk, potentially allowing their condition to worsen over time.

A second limitation stems from the **mismatch between training data and deployment context**. The NLP model is trained on Reddit and Twitter data which primarily consists of informal, public-facing social media expressions. In contrast, chatbot interactions in a counselling setting are more structured, context-driven and may involve more deliberate or guarded communication. Users may also switch between different tones, languages or levels of formality depending on the situation. This creates a generalisation challenge as the model may not perform consistently across these varied interaction styles. Furthermore, the chatbot environment may involve multi-turn conversations, nuanced context and even multilingual inputs (Singlish or code-switching), all of which extend beyond the scope of typical social media text. As such, the model's effectiveness in real-world deployment may be limited by its inability to fully adapt to these richer and more complex communication patterns.

Furthermore, with respect to the chosen NLP classifier, **more should be done to enhance model explainability**. For our project, we had prioritised establishing a functional classification pipeline as a foundation. To gain a strong preliminary understanding of the model's behaviour, we carried out an error analysis of its misclassifications. Indeed, it has suggested possible shortcomings of the model's decision-making process – such as the conflation of death-related vocabulary with high severity. However, to verify the validity of these limitations, more rigorous explainability techniques are required to trace predictions back to specific input features. In other words, while our current analysis provides useful insights, it may not achieve a fully transparent or definitive explanation of the model's internal reasoning.

Finally, the system has **not been piloted with real users** which introduces significant uncertainty regarding its real-world performance and usability. While the model may demonstrate promising results in controlled or experimental settings, actual user behaviour can differ substantially. Factors such as user trust, engagement levels, response honesty and interaction patterns are difficult to simulate without deployment. Additionally, scaling the system from a small test group in NUS to a broader population such as all Singaporeans aged 15 to 24 may surface unforeseen challenges, including increased variability in language use, differing cultural attitudes towards mental health and potential misuse of the system. Without real-world pilot data, it is difficult to assess how the model performs under practical conditions and whether it can maintain accuracy, fairness and user trust.

5.3 - Future Work

There are several improvements that could be made to the model in future. These include:

- Greater focus on **longitudinal PHQ tracking**, allowing the chatbot to monitor changes in a student's mental health over time. By establishing a personalised baseline for each student, the system could more accurately detect deviations from their typical mood patterns, improving early identification of at-risk individuals. This approach would also help differentiate between temporary stress fluctuations and sustained depressive trends, enhancing the reliability of alerts.
- Integrating the model with **UHC's electronic health records** (we do not have any records from UHC currently). This would be a critical step in bridging digital screening with formal clinical workflows. By linking PHQ responses and NLP insights to existing records, counsellors could access richer context such as prior consultations. This integration would support more informed decision-making, reduce duplicated assessments and allow for timely interventions.
- Conducting a more in-depth exploration of **explainability techniques**, so as to increase the transparency and interpretability of predictions made by the NLP classifier. Methods such as Local

Interpretable Model-Agnostic Explanations (LIME) or Shapley Additive Explanations (SHAP) could be applied to highlight the specific tokens or phrases that most influenced a given classification. This would enable counsellors to see not only the predicted severity level, but also the textual evidence behind it. Given BERT's attention-based architecture, we could consider attention maps as well. These would help us identify parts of the input text that the model weighted most heavily.

- Implementing a **pilot study with UHC** to provide a valuable real-world validation of the system's performance. By deploying the chatbot with a small, diverse cohort of students, researchers could evaluate engagement patterns, alert accuracy and overall user experience. Data collected from this pilot would inform adjustments to the model, refine thresholds for alerts and identify potential gaps or biases before broader implementation.
- **Expanding beyond depression to include anxiety and stress-related comorbidities** for future iterations. By incorporating validated screening tools for multiple mental health dimensions, the system could offer a more holistic view of student well-being. This expansion would allow for tailored interventions targeting overlapping symptoms and support a comprehensive mental health strategy across the university population.
- **Expanding the feature set of the current model by incorporating additional health-related metrics** such as heart rate, blood pressure and oxygen saturation could enhance its ability to assess depression severity. At present, the model relies solely on two inputs: PHQ-2 scores and users' free-text responses to chatbot prompts. While these features provide useful signals, they may be insufficient for a comprehensive and robust evaluation of an individual's mental health.

For the last point on expanding the feature set of the current model, we have incorporated and built a **health tracking dashboard** in Figma, where users can monitor a more comprehensive range of well-being indicators. While PHQ-2 scores are tracked over time to assess depressive symptom severity, the dashboard also allows users to optionally log additional health metrics correlated with depression risk such as blood pressure, heart rate, breathing rate and sleeping hours. These inputs are entirely voluntary, ensuring that the system enhances rather than burdens the user experience.

By aggregating both psychological signals (PHQ scores and free-text responses) and physiological data, the platform constructs a more holistic representation of each student's health status. This addresses the limitations of relying on a narrow feature set, strengthening the model's ability to detect subtle patterns of deterioration. Consequently, the system can generate more robust, clinically grounded signals for the triage engine while also empowering students with a comprehensive health report that supports self-awareness and more informed discussions with counsellors during scheduled sessions.

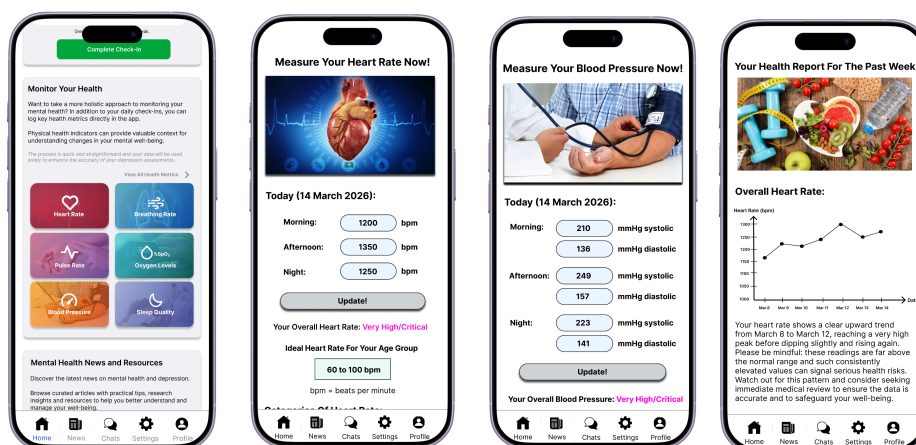


Image 1: Monitor Your Health section in the home page of the app, allowing users to monitor and track their health over time

Image 2: One example of health metric included in the app - heart rate (gives ideal figures for age range and suggestions to improve)

Image 3: Another example of health metric included in the app - blood pressure (gives ideal figures for age range and suggestions to improve)

Image 4: Overall health report page, present health status trends for each category in graphs with short descriptions of how figures fare

Appendix:

Relevant Links

- Codebase
 - [GitHub repository](#)
- Prototyping
 - [NUS Depression App Design – Figma \(Design View\)](#)
 - [NUS Depression App Design – Figma \(Prototype View\)](#)
 - [Code demonstration of dynamic questioning in chatbot](#)
- Fine-tuning BERT
 - [Original Reddit and Twitter dataset \(HelaDepDet\)](#)
 - [Cleaned dataset for fine-tuning](#)
 - [Model weights for fine-tuned classification model](#)
 - [Model weights for fine-tuned regression model](#)
- Group Presentation
 - [Presentation Slides](#)